

Project Report

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Horus: A deeply trained model to identify and remove targeted universal adversarial patches

1 Abstract

Object classification has always been a central focus in computer vision research. The primary task is to identify the object within an image. State-of-the-art classification methods, such as convolutional neural networks and deep neural networks, have achieved remarkable results. However, these models are trained as black boxes and can recognize patterns that are often counter-intuitive to human observers. This vulnerability has given rise to a class of attacks called "adversarial patching." In this attack, a specifically crafted pattern is embedded into an image to deceive a recognition model into making incorrect classifications. To address this challenge, we are proposing the development of Horus, a preprocessing model designed to identify and remove local adversarial patches from images before the classification process. It detects the targeted adversarial patches and covers them by adding a monochromatic mask over them. This helps in diminishing the effects of the patch and lowering the rate of misclassifications.

Keywords: Adversarial patch, Targeted adversarial attack, CNN, image classification, adversarial patch detection

2 Introduction

Adversarial patches are a type of attack in computer vision and machine learning. In this attack, a small patch is added to an image which results in its mis-classification by the model. Brown et al [1] introduces a way of creating universal and perceptible adversarial patches that are image-independent. This allows the attacker to use the patch to attack a wide range of images, and 'push' the model to predict an incorrect target class, without having any prior information about the image. Our objective is to significantly reduce the impact of these patches on black box attacks, while also not compromising

with the correct classification of the images. In recent years, various approaches have been introduced to tackle this issue. We have reviewed a survey on these existing methods [2].

3 Methodology

3.1 Datasets

For our model, we have used a downsampled subset of the Imagenet dataset. This subset contains 10000 annotated images categorised into one of the 1000 labels, thus each category having 10 samples. We have randomly selected 8500 images for training and 1500 for testing. Every image in this subset has the (H, W, C) dimensions of $(224, 224, 3)$.

3.2 Models

For the image classification of the the Imagenet samples, we used a model of the ResNet34 architecture which was pre-trained on larger Imagenet dataset.

Due to the lack of properly annotated adversarial patches on the Imagenet classes, we trained a model to create targeted attacks based on [1]. In this paper, the authors create their attack by completely replacing a part of the image with their patch. They train their patch over a variety of images, applying a random shape, translation, scaling, and rotation on the patch in each image, optimizing using gradient descent . For simplicity, in our training we used, a square patch of given sizes and without any rotation. We start with a random noise patch and while training we provide a target class, for which we want to increase the prediction confidence using the patch, calculate the classification loss, and then use gradient descent with momentum to optimize the patch. Eventually the patches start looking like the labels of the target class.

For controlling the adversarial effects of the patch, we need to first detect the adversarial patches. Jiang Liu et al [3] propose Segment and Complete defense (SAC), a model for defending object detectors against such patch attacks by detecting and removing the patches. They use image segmentation to identify the pixels belonging to the patch and use adversarial training with pre-generated adversarial examples to train the segmentation, and use shape completion model to give a square or rectangular region. We used their implementation to help identifying the pixels related to the patches, and then try to lower the misclassifications caused by patches, by covering the segmented area with an opaque mask, thus hiding the patch. Though this creates a new monochromatic patch over the original adversarial one, but this new patch does not 'imitate' any target class and has much lesser salient features. This will help the model to 'focus' on the the other regions of the image that could help in correct classification.

4 Evaluation

There are two aspects of the model that we need to evaluate: model classification accuracy and the patch detection accuracy.

4.1 Classification Accuracy

We test the classification accuracy of the Resnet36 model by finding the class labels with the highest confidence score and compare it with the true labels. We have named it top-1 accuracy where ‘1’ refers to the number of classes in the results that we are comparing. Since the Imagenet dataset has a wide range of classes and images might have objects of multiple labels, top-5 accuracy is also widely used for evaluation. We consider a prediction to be Top-5 accurate if the true label is among the five predicted labels with highest probabilities [4].

$$Top - 1 Accuracy = \frac{samples(Predicted == True)}{samples(Total)} \quad (1)$$

$$Top - 5 Accuracy = \frac{samples(True \in Top 5 Predicted Classes)}{samples(Total)} \quad (2)$$

4.2 Patch Detection Accuracy

Since we did not have an annotated ‘image-with-patch’ dataset with class labels and boundary labels for the patches, we created the patches and placed the patch on the image in five different locations. Since we know the boundary labels of the patches which are added during the test time, we can use them for finding the accuracy of the patch mask.

For evaluating the performance of the patch detection, we use Intersection over Union (IoU) metric [5]. We count the number of pixels along any one channel that are present in both the patch and the mask (for the intersection) and the total number of pixels covered by patch and mask combined (for the union). If for some image, the detector was unable to find any patch, then we take the IoU value to be zero. Then we take the average for all the five orientations of the patches to find the average IoU value.

5 Results

The Resnet36 model was able to achieve a top-1 accuracy of about 81% and top-5 accuracy of over 96% on the image samples without the patches.

After creating the patches, we also tested the classification accuracies with the patches of target labels ‘goldfish’. For the ‘goldfish’ label with the adversarial patch of size (48p X 48p), we got top-1 and top-5 accuracies of 31.31% and 74.69% respectively. After using the patch detector and applying the mask, we could obtain average top-1 accuracy of 57.53% and top-5 accuracy of 84.06% (Appendix Figure 1)

We calculated the average IoU by adding the IoU for individual predictions and dividing by the number of test samples. We received an average IoU of 41.36%. (Appendix Figure 1)

6 Conclusions and Future Work

By observing the results, we can clearly see a significant increase in both the top-1 and top-5 classification accuracies of the model after adding the patch defence mask. The targeted adversarial patches

[1] are optimized to ‘fool’ the model to predict the given images as the target classes. This primarily works by ‘amplifying’ the salient features of the patch, making it more significant for prediction. As a result, the model ‘overlooks’ or gives less importance to the other regions of the image while making predictions. The model can be fooled even when the patch is not placed on the object of importance in the image.

By adding a mask over the patch, we can reduce the effect of the patch by decreasing its salience greatly. This will make the model give weightage to the surroundings of the patch, which helps in making correct predictions.

However there are some issues with the patch mask, which can cause some misclassifications when applied on the image. Since the mask increases the significance of the background, if the patch covers most of the pixels corresponding to the object of interest (here, object of interest refers to the object in the image that corresponds to the its true label), then model will not be able to make correct predictions only based on the surrounding and the patch. Also, since patch mask may also become inadvertently an object of interest (thus behaving like another adversarial patch), this may lead to some highly erroneous predictions.

Instead of removing the suspicious patch region or adding a monochromatic featureless mask over it, we can try denoising the patch region by averaging the pixels to reduce frequency. This will be helpful in some scenarios where our mask struggled. For example, if an image has a bird flying in the sky and there was a patch covering the bird, then the model was unable to make the correct classification despite being able to detect the patch. If we use some form of averaging, such that the region of interest is filled with pixel values having the color of the bird, the confidence score for the true label might have increased. We can also test and compare the effects of different patch sizes, target classes and orientation, and test the efficacy of the mask, which will help generalize the results.

7 References

- [1] T. B. Brown, D. Mane, A. Roy, M. Abadi, and J. Gilmer, “Adversarial patch,” in NIPS Workshop, 2017.
- [2] A. Sharma, Y. Bian, P. Munz, and A. Narayan, “Adversarial patch attacks and defences in vision-based tasks: A survey,” arXiv preprint arXiv:2206.08304, 2022
- [3] Jiang Liu, Alexander Levine, Chun Pong Lau, Rama Chellappa, and Soheil Feizi. 2021. Segment and Complete: Defending Object Detectors against Adversarial Patch Attacks with Robust Patch Detection. arXiv:2112.04532 (2021)
- [4] <https://towardsdatascience.com/accuracy-and-loss-things-to-know-about-the-top-1-and-top-5-accuracy-1d6beb8f6df3>
- [5] <https://learnopencv.com/intersection-over-union-iou-in-object-detection-and-segmentation/>

APPENDIX

1. Evaluation Results:

Table 1) Evaluation Results

	Top-1 Accuracy	Top-5 Accuracy	Average IoU
Without Patch	81	96	N/A
With Patch	31.31	74.69	N/A
With Patch and Mask	57.53	84.06	41

Figure 1) Evaluation metrics values after testing 1500 test sample

Top_1 accuracy - 0.3131127450980392
Top_5 accuracy - 0.7469362745098039
Top_1 patch accuracy - 0.5753676470588235
Top_5 patch accuracy - 0.8406862745098039
Average IOU - 0.4136657140420354

2. Adversarial Patches Examples:

Figure 2.1) Toaster and Goldfish Adversarial Patches of size (32x32)

toaster, size 32



goldfish, size 32

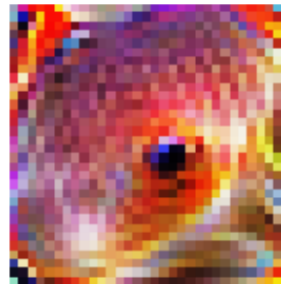


Figure 2.2) Toaster and Goldfish Adversarial Patches of size (48x48)

toaster, size 48



goldfish, size 48



3. Test Results Examples (labels above the images are true classes):

Figures 3.1.1 to 3.1.4) Examples with correct Top-1 prediction after applying defence mask

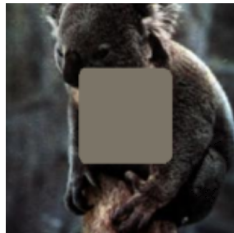
Predictions -
goldfish 86.8980
axolotl 8.2017
rock beauty 0.8662
tench 0.5873
common newt 0.2769

koala



Predictions -
koala 44.2195
indri 28.0545
marmoset 4.5702
titi 3.4228
guenon 3.2634

koala



Predictions -
goldfish 52.5539
chiton 32.9421
sea slug 1.5069
flatworm 1.0925
isopod 1.0685

chiton



Predictions -
chiton 97.2102
trilobite 1.9588
isopod 0.2791
scorpion 0.1047
flatworm 0.0385

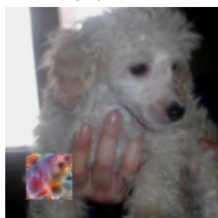
chiton



Figures 3.2.1 to 3.2.2) Example where the Top-1 prediction remains same but the confidence increases

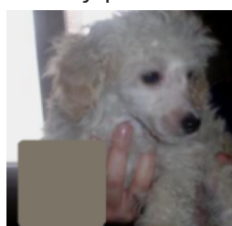
Predictions -
toy poodle 39.5232
miniature poodle 36.9518
standard poodle 9.0901
goldfish 7.3784
kuvasz 1.2065

toy poodle



Predictions -
toy poodle 54.7258
miniature poodle 44.8163
standard poodle 0.3033
Maltese dog 0.0935
kuvasz 0.0186

toy poodle



Figures 3.3.1 to 3.3.2) Example where patch detection was incorrect

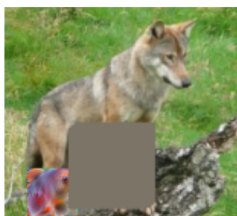
Predictions -
goldfish 21.1893
red wolf 10.5523
timber wolf 10.0696
dingo 7.0948
coyote 6.0110

timber wolf



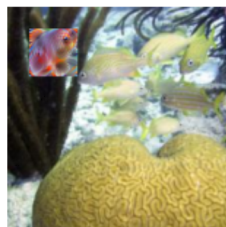
Predictions -
timber wolf 21.4059
goldfish 20.6897
red wolf 11.6690
coyote 7.3984
white wolf 4.5541

timber wolf



Predictions -
brain coral 94.2993
goldfish 3.3952
coral reef 2.1307
rock beauty 0.0673
anemone fish 0.0497

brain coral



Predictions -
brain coral 98.2518
goldfish 0.9500
coral reef 0.6698
anemone fish 0.0452
rock beauty 0.0373

brain coral

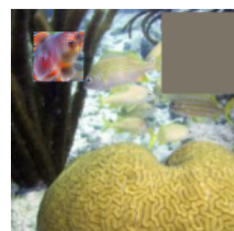
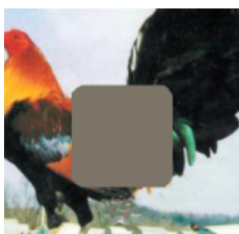


Figure 3.4.1 to 3.4.2) Example where patch detection is correct but Top-1 prediction is incorrect

Predictions - goldfish 99.9913 tench 0.0023 gar 0.0007 coho 0.0005 tailed frog 0.0005 goldfish 	Predictions - lakeside 17.7232 goldfish 4.9143 boathouse 3.3436 vizsla 3.1259 pot 2.6693 goldfish 	Predictions - goldfish 98.4250 rock beauty 0.1485 kite 0.1109 macaw 0.1083 lorikeet 0.0712 cock 	Predictions - black grouse 54.5918 toucan 4.2736 lorikeet 4.1720 switch 2.5195 joystick 2.0800 cock 
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